Exercise Sheet

Lecture 1

Vector Spaces and Subspaces

Unless stated otherwise, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$ and core problems are finite dimensional. Displayed function and sequence spaces are examples or bridges. Tags in parentheses indicate the flavour of a problem; a $\star$ marks a harder one.

Core tutorial route

Work **1.1, 1.6, 1.8, 1.10, 1.18, 1.19, 1.24, and 1.25** in that order (about 65–80 minutes before discussion). This eight-problem route checks calculation, an axiom diagnosis, polynomial/sequence/function/matrix examples, scalar-field dependence, and both computational and explanatory direct-sum work.

Additional practice: 1.2–1.5, 1.7, 1.9, 1.11, 1.13–1.17, 1.21–1.23, 1.27. **Bridge:** 1.12 (ODE solution spaces). **Extension:** 1.20 and 1.26.

Warm-up and axioms

Exercise 1.1 (computational) In $\mathbb{C}^2$, let $u = (1+i,\ 2)$ and $w = (3,\ -i)$. Compute $u+w$, the vector $(2-i)\,u$, and the additive inverse $-u$.

Answer. $u+w = (4+i,\ 2-i)$; $(2-i)u = (3+i,\ 4-2i)$; $-u = (-1-i,\ -2)$.

Exercise 1.2 (computational) In $\mathbb{R}^3$, let $u = (2,-1,3)$ and $w = (0,4,-2)$. Compute $u+w$, $3u$, and $2u-w$.

Answer. $u+w = (2,3,1)$; $3u = (6,-3,9)$; $2u-w = (4,-6,8)$.

Exercise 1.3 (computational) In $\mathcal{P}_3(\mathbb{R})$, let $p = 2 - t + 3t^2$ and $q = 1 + 4t - t^2 + t^3$. Compute $p+q$, $3p$, and $2p-q$.

Answer. $p+q = 3+3t+2t^2+t^3$; $3p = 6-3t+9t^2$; $2p-q = 3-6t+7t^2-t^3$.

Exercise 1.4 (computational) In $M_{2\times 2}(\mathbb{R})$, let $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & -1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 3 & -1 \\ 4 & 2 \end{smallmatrix}\right)$. Compute $A+B$, $2A$, and $A-2B$.

Answer. $A+B = \left(\begin{smallmatrix} 4 & 1 \\ 4 & 1 \end{smallmatrix}\right)$; $2A = \left(\begin{smallmatrix} 2 & 4 \\ 0 & -2 \end{smallmatrix}\right)$; $A-2B = \left(\begin{smallmatrix} -5 & 4 \\ -8 & -5 \end{smallmatrix}\right)$.

Exercise 1.5 (proof) Using only the vector space axioms, prove that $a\,v = 0$ (with $a \in \mathbb{F}$, $v \in V$) implies $a = 0$ or $v = 0$.

Hint. Suppose $a \neq 0$. The field then hands you a^{-1}; multiply the equation by it and read off v.

Exercise 1.6 (proof) Decide, with proof, whether $\mathbb{R}^2$ with the *usual* addition but the scalar multiplication $a \odot (x,y) := (ax,\ 0)$ is a vector space over $\mathbb{R}$.

Hint. One axiom is enough to settle it — test the scalar 1 on a vector whose second coordinate is non-zero.

Exercise 1.7 (proof) Prove the cancellation law in a vector space: if $u, v, w \in V$ satisfy $u + w = v + w$, then $u = v$. Use only the axioms.

Hint. w has an additive inverse. Add it to both sides and let associativity do the rest.

Recognising vector spaces

Exercise 1.8 (computational) Which of the following are vector spaces under the natural operations?

1. the polynomials in $\mathcal{P}(\mathbb{R})$ of degree *exactly* 3;
2. the polynomials $p \in \mathcal{P}_3(\mathbb{R})$ with $p(1) = 0$;
3. the real sequences (x_n) with $x_n \to 0$.

Answer. (1) No (not closed under +; also lacks 0). (2) Yes. (3) Yes.

Exercise 1.9 (computational) Which of the following subsets of $\mathbb{R}^2$ are vector spaces under the usual operations?

1. $\{(x, y) : 2x - 3y = 0\}$;
2. $\{(x, y) : x \geq 0\}$;
3. $\{(x, y) : xy = 0\}$.

Answer. (1) Yes. (2) No (not closed under scaling by -1). (3) No (not closed under +).

Exercise 1.10 (computational) Which of the following subsets of $\mathbb{R}^{\mathbb{R}}$ (the real functions of one real variable) are vector spaces?

1. $\{f : f(3) = 0\}$;
2. $\{f : f(3) = 1\}$;
3. $\{f : f \text{ is bounded}\}$.

Answer. (1) Yes. (2) No (zero function fails $f(3) = 1$). (3) Yes.

Exercise 1.11 (computational) Which of the following subsets of the space of real sequences (x_n) are vector spaces?

1. $\{(x_n) : x_1 = x_2\}$;
2. $\{(x_n) : \text{only finitely many } x_n \text{ are nonzero}\}$;
3. $\{(x_n) : x_n \to 5\}$.

Answer. (1) Yes. (2) Yes. (3) No (zero sequence converges to $0 \neq 5$).

Exercise 1.12 (proof) Inside the vector space of twice-differentiable functions $y\colon \mathbb{R} \to \mathbb{R}$, show that the solutions of $\ddot{y} + 5\dot{y} + 6y = 0$ form a vector space, but the solutions of $\ddot{y} + 5\dot{y} + 6y = t$ do not.

Hint. Differentiation is linear, so check closure for the homogeneous equation. For the inhomogeneous one, ask whether the zero function is a solution.

Exercise 1.13 (proof) Prove that $\{p \in \mathcal{P}(\mathbb{R}) : p(1) = p(2)\}$ is a subspace of $\mathcal{P}(\mathbb{R})$, but that $\{p \in \mathcal{P}(\mathbb{R}) : p(1)\,p(2) = 0\}$ is not.

Hint. $p(1) = p(2)$ is a homogeneous linear condition on p; $p(1)\,p(2) = 0$ is not linear at all. For the second set, find two polynomials in it whose sum is not.

Subspaces

Exercise 1.14 (computational) For each subset of $\mathbb{F}^3$, decide whether it is a subspace.

1. $\{(x_1, x_2, x_3) : x_1 + 2x_2 + 3x_3 = 0\}$;

2. $\{(x_1, x_2, x_3) : x_1 + 2x_2 + 3x_3 = 4\}$;
3. $\{(x_1, x_2, x_3) : x_1x_2x_3 = 0\}$.

Answer. (1) Subspace. (2) Not a subspace (0 fails the condition). (3) Not a subspace (not closed under addition).

Exercise 1.15 (computational) For each subset of $\mathbb{R}^3$, decide whether it is a subspace.

1. $\{(x, y, z) : 2x - y = 0 \text{ and } z = 0\}$;
2. $\{(x, y, z) : x^2 = y^2\}$;
3. $\{(t, 2t, 3t) : t \in \mathbb{R}\}$.

Answer. (1) Subspace. (2) Not a subspace. (3) Subspace.

Exercise 1.16 (computational) For each subset of $\mathbb{R}^4$, decide whether it is a subspace.

1. $\{(x_1, x_2, x_3, x_4) : x_1 - x_2 + x_3 - x_4 = 0\}$;
2. $\{(x_1, x_2, x_3, x_4) : x_1 = x_4\}$;
3. $\{(x_1, x_2, x_3, x_4) : x_1 + x_2 = 1\}$.

Answer. (1) Subspace. (2) Subspace. (3) Not a subspace (0 fails the condition).

Exercise 1.17 (computational) Decide whether each set is a subspace of the indicated space.

1. $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$ in $\mathcal{P}_3(\mathbb{R})$;
2. $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 1\}$ in $\mathcal{P}_3(\mathbb{R})$;
3. $\{A \in M_{2\times2}(\mathbb{R}) : a_{11} = a_{22}\}$ in $M_{2\times2}(\mathbb{R})$.

Answer. (1) Subspace. (2) Not a subspace (0 fails). (3) Subspace.

Exercise 1.18 (proof) Prove that the trace-zero matrices $\{A \in M_{n\times n}(\mathbb{F}) : \operatorname{tr} A = 0\}$ form a subspace of $M_{n\times n}(\mathbb{F})$, and that the invertible matrices do *not*.

Hint. The trace is linear, which settles the first set at once. For the invertible matrices, look at the zero matrix — or at $I + (-I)$.

Exercise 1.19 ($\star$) Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$? Is it a subspace of $\mathbb{C}^2$ regarded as a *real* vector space?

Hint. The deciding operation is multiplication by the complex scalar i: check whether $i \cdot (1, 0)$ stays in $\mathbb{R}^2$.

Exercise 1.20 (proof) Let U and W be subspaces of a vector space V. Prove that $U \cup W$ is a subspace of V if and only if $U \subseteq W$ or $W \subseteq U$.

Hint. One direction is immediate. For the other, argue by contradiction: if neither contains the other, pick $u \in U \setminus W$ and $w \in W \setminus U$ and examine $u + w$.

Sums and direct sums

Exercise 1.21 (computational) In $M_{2\times2}(\mathbb{R})$, write $A = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$ as the sum of a symmetric and an antisymmetric matrix.

Answer. $A = \begin{pmatrix} 4 & 4 \\ 4 & 2 \end{pmatrix} + \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}$.

Exercise 1.22 (computational) In $M_{2\times2}(\mathbb{R})$, write $A = \begin{pmatrix} 3 & 5 \\ 1 & 7 \end{pmatrix}$ as the sum of a symmetric and an antisymmetric matrix.

Answer. $A = \begin{pmatrix} 3 & 3 \\ 3 & 7 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$.

Exercise 1.23 (computational) In $M_{3\times 3}(\mathbb{R})$, write $A = \begin{pmatrix} 2 & 4 & 0 \\ 0 & 2 & 6 \\ 4 & -2 & 2 \end{pmatrix}$ as the sum of a symmetric and an antisymmetric matrix.

Answer. $A = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 2 & -2 \\ -2 & 0 & 4 \\ 2 & -4 & 0 \end{pmatrix}$.

Exercise 1.24 (computational) In $\mathbb{R}^3$, for each pair of subspaces decide whether the sum $U+W$ is direct.

1. $U = \{(x, y, 0)\}$ and $W = \{(0, 0, z)\}$;
2. $U = \{(x, y, 0)\}$ and $W = \{(x, 0, z)\}$.

Answer. (1) Direct ($U \cap W = \{0\}$, and $U \oplus W = \mathbb{R}^3$). (2) Not direct ($U \cap W = \{(x, 0, 0)\} \neq \{0\}$).

Exercise 1.25 (proof) Let U_{e} and U_{o} be the even and odd functions in $\mathbb{R}^{\mathbb{R}}$ (so $f(-x) = f(x)$ and $g(-x) = -g(x)$, respectively). Show that $\mathbb{R}^{\mathbb{R}} = U_{\mathrm{e}} \oplus U_{\mathrm{o}}$.

Hint. Every f splits as $\frac{1}{2}\big(f(x) + f(-x)\big) + \frac{1}{2}\big(f(x) - f(-x)\big)$; check what each half is. For directness, ask what a function that is both even and odd can be.

Exercise 1.26 ($\star$) Give three subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ such that $V_i \cap V_j = \{0\}$ for every $i \neq j$, yet $V_1 + V_2 + V_3$ is *not* direct. (This shows pairwise-trivial intersections are not enough for three or more subspaces.)

Hint. Pairwise-trivial intersection is weaker than directness — try three distinct lines lying in a single plane through the origin.

Exercise 1.27 (proof) Let U and W be subspaces of a vector space V. Prove that $U + W = W$ if and only if $U \subseteq W$.

Hint. Both inclusions are short. For $U + W \subseteq W$ use closure of W; for the converse write $u = u + 0$.